

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct

I **MOHAMMED NAUASIN** (Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1.5	1.5	1.5	1.5	1.5
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1/2	1/2	1/2	1/2	1/2

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::f000:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::f000:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Assessment-3

1. Debugging IPv4 Configuration Given

IP Address 192.168.10.65
Subnet Mask 255.255.255.192 (/26)
Default gateway 192.168.10.1

a) Subnet

192.168.10.64/26

b) Network Address

192.168.10.64

c) Broadcast Address

192.168.10.127

d) Valid Host Range

192.168.10.25 - 192.168.10.26

e) Verification

IP Address → Valid

Default gateway → Invalid (gateway 192.168.10.1)
belongs to another subnet

f) Configuration Error

Incorrect default gateway

g) Optimized Configuration

IP Address 192.168.10.65

Subnet Mask 255.255.255.192

Default gateway 192.168.10.65 (or another valid router

IP in this Subnet such as 192.168.10.126

h) Usable Hosts

62

2. IPV6 Debugging

Device A

2001::db8::4400:42:8329/64

Expanded

2001:0db8:0000:0000:0000:4400:0042:8329

Network prefix

2001:db8:0:0::/64

Device B

2001:db8:0:0:1::/64

Expanded

2001:0db8:000:0001:0000:0000:0000:0001

Network prefix

2001:db8:0:0:1::/64

Same Subnet?

No

Mismatch

Different network prefixes

Optimized Configuration

Assign Device B an address such as

2001:db8::1/64

Now both device are in:

2001:db8::0/64

Available Host addresses

$$2^{64} = 18,446,744,073,709,551,616$$

3 Subnet optimization

Network

192.168.1.0/24

/26

Network: 192.168.1.0

Broadcast: 192.168.1.63

Host Range: 192.168.1.1 - 192.168.1.62

Usable Host 62

/27

Network: 192.168.1.64

Broadcast: 192.168.1.95

Host Range: 192.168.1.65 - 192.168.1.94

Usable Hosts 30

/28

Network: 192.168.1.96

Broadcast 192.168.1.111

Host Range 192.168.1.97-192.168.1.110

Usable Hosts 14

Unused Addresses

/26 → 2

/27 → 2

/28 → 2

Inefficient Allocation

Some department have more IPs than required while others have insufficient IPs.

Debugging

Fixed size subnet allocation causes address wastage

Recommendation

Use VSM (Variable length Subnet Mask) to allocate Subnet sizes based on actual department requirements

4 IPv6 Debugging

Device A

Expanded

2001:db8:0000:0000:0000:4000:0042:8329

Network Prefix

2001:db8:0:0::/64

Device B

Expanded

2001:db8:0000:000:0001:0000:000:0001

Network Prefix

2001:db8:0:0:1::1/64

Same Subnet
No

Mismatch

Different prefixes

Correlated Address

2001:db8::1/64

Available Hosts

$2^{64} = 18,446,744,073,709,551,616$

5. Routing Table Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.2
192.168.2.0/24	10.0.3
Default Route	10.0.1

Destination IP

192.168.2.100

Matching Route

192.168.2.0/24

Next-Hop Router

10.0.3

Destination Verification

Yes, 192.168.2.100 belongs to 192.168.2.0/24

Configuration Errors

No routing error. The pack should be forwarded through 10.0.3

optimization

Keep specific routes before the default route and ensure routing entries are updated

Route used if No Match Exists

Default Route $\rightarrow$ 10.0.1